

AI Health Symptom Assessment Agent:

```
<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8" />

<meta name="viewport" content="width=device-width, initial-scale=1.0" />

<title>AI Health Symptom Assessment Agent</title>

<style>

  *, *::before, *::after { box-sizing: border-box; margin: 0; padding: 0; }

  :root {

    --bg: #ffffff;

    --surface: #f7f8fa;

    --border: #e5e7eb;

    --text: #1f2328;

    --muted: #57606a;

    --accent: #3b82d4;

    --accent-dark: #2563b5;

    --secondary: #7c5cd8;

    --danger: #dc2626;

    --warning: #d97706;

    --success: #16a34a;

    --info: #0891b2;

    --radius: 8px;

    --shadow: 0 1px 3px rgba(0,0,0,0.08), 0 1px 2px rgba(0,0,0,0.04);
```

```
}
```

```
body {
```

```
  font-family: -apple-system, "Segoe UI", system-ui, sans-serif;
```

```
  font-size: 14px;
```

```
  line-height: 1.6;
```

```
  background: #eef1f6;
```

```
  color: var(--text);
```

```
  min-height: 100vh;
```

```
}
```

```
/* — HEADER — */
```

```
.header {
```

```
  background: var(--text);
```

```
  color: #fff;
```

```
  padding: 0 24px;
```

```
  height: 56px;
```

```
  display: flex;
```

```
  align-items: center;
```

```
  justify-content: space-between;
```

```
  position: sticky;
```

```
  top: 0;
```

```
  z-index: 100;
```

```
}
```

```
.header-brand {
```

```
display: flex;

align-items: center;

gap: 10px;

font-weight: 600;

font-size: 15px;

letter-spacing: 0.01em;
}

.header-brand svg { flex-shrink: 0; }

.header-badge {

background: var(--accent);

color: #fff;

font-size: 10px;

font-weight: 700;

padding: 2px 7px;

border-radius: 20px;

letter-spacing: 0.05em;

text-transform: uppercase;
}

.header-meta {

font-size: 11px;

color: #9ca3af;

text-align: right;
}

/* — LAYOUT — */
```

```
.layout {  
  max-width: 1100px;  
  margin: 0 auto;  
  padding: 24px 16px 48px;  
  display: grid;  
  grid-template-columns: 340px 1fr;  
  gap: 20px;  
  align-items: start;  
}
```

```
@media (max-width: 800px) {  
  .layout { grid-template-columns: 1fr; }  
}
```

```
/* — CARDS — */
```

```
.card {  
  background: var(--bg);  
  border: 1px solid var(--border);  
  border-radius: var(--radius);  
  box-shadow: var(--shadow);  
  overflow: hidden;  
}  
  
.card-header {  
  background: var(--surface);  
  border-bottom: 1px solid var(--border);
```

```
padding: 14px 18px;

display: flex;

align-items: center;

gap: 8px;

font-weight: 600;

font-size: 13px;

color: var(--text);

text-transform: uppercase;

letter-spacing: 0.04em;

}

.card-header-dot {

width: 8px; height: 8px;

border-radius: 50%;

flex-shrink: 0;

}

.card-body { padding: 18px; }
```

```
/* — FORM ELEMENTS — */
```

```
label {

display: block;

font-size: 12px;

font-weight: 600;

color: var(--muted);

text-transform: uppercase;

letter-spacing: 0.04em;
```

```
margin-bottom: 6px;

}

input[type="text"],
input[type="number"],
select,
textarea {

width: 100%;

padding: 9px 12px;

border: 1px solid var(--border);

border-radius: 6px;

font-size: 13.5px;

font-family: inherit;

color: var(--text);

background: var(--bg);

transition: border-color 0.15s;

outline: none;

appearance: none;

}

input:focus, select:focus, textarea:focus {

border-color: var(--accent);

box-shadow: 0 0 3px rgba(59,130,212,0.12);

}

textarea { resize: vertical; min-height: 90px; }

.form-row { margin-bottom: 14px; }
```

```
.form-grid-2 { display: grid; grid-template-columns: 1fr 1fr; gap: 12px; }
```

```
/* — SYMPTOM TAGS — */
```

```
.tag-input-wrap {
```

```
  display: flex;
```

```
  gap: 6px;
```

```
  margin-bottom: 8px;
```

```
}
```

```
.tag-input-wrap input { flex: 1; }
```

```
.btn-tag-add {
```

```
  padding: 9px 14px;
```

```
  background: var(--surface);
```

```
  border: 1px solid var(--border);
```

```
  border-radius: 6px;
```

```
  font-size: 18px;
```

```
  cursor: pointer;
```

```
  color: var(--accent);
```

```
  font-weight: 700;
```

```
  line-height: 1;
```

```
  transition: background 0.15s;
```

```
}
```

```
.btn-tag-add:hover { background: #e8edf5; }
```

```
.tags-container {
```

```
  display: flex;
```

```
flex-wrap: wrap;

gap: 6px;

min-height: 32px;

padding: 4px 0;
}

.tag {

display: inline-flex;

align-items: center;

gap: 5px;

background: #dbeafe;

color: #1e40af;

border: 1px solid #bfdbfe;

border-radius: 20px;

padding: 3px 10px 3px 12px;

font-size: 12px;

font-weight: 500;
}

.tag-remove {

cursor: pointer;

font-size: 14px;

line-height: 1;

color: #6b7280;

background: none;

border: none;

padding: 0;
```

```
margin-left: 2px;

}

.tag-remove:hover { color: var(--danger); }
```

```
.quick-symptoms {

display: flex;

flex-wrap: wrap;

gap: 5px;

margin-top: 8px;

}
```

```
.quick-tag {

background: var(--surface);

border: 1px solid var(--border);

border-radius: 20px;

padding: 3px 10px;

font-size: 11.5px;

cursor: pointer;

color: var(--muted);

transition: all 0.15s;

}
```

```
.quick-tag:hover {

background: #dbeafe;

color: #1e40af;

border-color: #bfdbfe;

}
```

```
/* — BUTTONS — */

.btn {

  display: inline-flex;

  align-items: center;

  justify-content: center;

  gap: 7px;

  padding: 10px 20px;

  border-radius: 7px;

  font-size: 13.5px;

  font-weight: 600;

  font-family: inherit;

  cursor: pointer;

  border: none;

  transition: all 0.15s;

  letter-spacing: 0.01em;

}

.btn-primary {

  background: var(--accent);

  color: #fff;

  width: 100%;

}

.btn-primary:hover:not(:disabled) { background: var(--accent-dark); }

.btn-primary:disabled {

  background: #9ca3af;
```

```
    cursor: not-allowed;
}

.btn-ghost {
    background: var(--surface);
    color: var(--muted);
    border: 1px solid var(--border);
}

.btn-ghost:hover { background: #eaecf0; color: var(--text); }
```

```
.btn-row {
    display: flex;
    gap: 8px;
    margin-top: 8px;
}
```

```
.btn-row .btn { flex: 1; }
```

```
/* — SPINNER — */
```

```
.spinner {
    width: 16px; height: 16px;
    border: 2px solid rgba(255,255,255,0.4);
    border-top-color: #fff;
    border-radius: 50%;
    animation: spin 0.7s linear infinite;
    display: none;
}
```

```
@keyframes spin { to { transform: rotate(360deg); } }
```

```
.loading .spinner { display: block; }
```

```
.loading .btn-label { display: none; }
```

```
/* — CHAT PANEL — */
```

```
.chat-panel { display: flex; flex-direction: column; gap: 16px; }
```

```
/* — WELCOME STATE — */
```

```
.welcome-box {
```

```
  background: linear-gradient(135deg, #f0f7ff 0%, #f5f0ff 100%);
```

```
  border: 1px solid #c7d9f5;
```

```
  border-radius: var(--radius);
```

```
  padding: 32px 24px;
```

```
  text-align: center;
```

```
}
```

```
.welcome-icon {
```

```
  width: 56px; height: 56px;
```

```
  background: var(--accent);
```

```
  border-radius: 50%;
```

```
  display: flex;
```

```
  align-items: center;
```

```
  justify-content: center;
```

```
  margin: 0 auto 16px;
```

```
}
```

```
.welcome-box h2 { font-size: 18px; margin-bottom: 8px; color: var(--text); }
```

```
.welcome-box p { color: var(--muted); font-size: 13px; max-width: 380px; margin: 0 auto 20px; }
```

```
.feature-grid {  
  display: grid;  
  grid-template-columns: 1fr 1fr;  
  gap: 10px;  
  text-align: left;  
}
```

```
.feature-item {  
  background: #fff;  
  border: 1px solid var(--border);  
  border-radius: 6px;  
  padding: 10px 12px;  
  font-size: 12px;  
  color: var(--muted);  
  display: flex;  
  align-items: flex-start;  
  gap: 8px;  
}
```

```
.feature-item-icon {  
  width: 22px; height: 22px;  
  border-radius: 4px;  
  flex-shrink: 0;  
  display: flex;  
  align-items: center;  
  justify-content: center;
```

```
font-size: 12px;

font-weight: 700;

color: #fff;

}
```

```
/* — ASSESSMENT RESULT — */
```

```
.assessment-card {

background: var(--bg);

border: 1px solid var(--border);

border-radius: var(--radius);

overflow: hidden;

box-shadow: var(--shadow);

}
```

```
.assessment-header {

padding: 16px 18px;

display: flex;

align-items: center;

justify-content: space-between;

border-bottom: 1px solid var(--border);

}
```

```
.assessment-title {

font-weight: 700;

font-size: 14px;

display: flex;

align-items: center;
```

```
    gap: 8px;
}

.assessment-timestamp {

    font-size: 11px;

    color: var(--muted);
}
```

```
/* Risk badge */

.risk-badge {

    display: inline-flex;

    align-items: center;

    gap: 5px;

    padding: 4px 12px;

    border-radius: 20px;

    font-size: 11.5px;

    font-weight: 700;

    text-transform: uppercase;

    letter-spacing: 0.06em;
}

.risk-low    { background: #dcfce7; color: #15803d; border: 1px solid #bbf7d0; }
.risk-moderate { background: #fef9c3; color: #a16207; border: 1px solid #fde68a; }
.risk-high    { background: #fee2e2; color: #b91c1c; border: 1px solid #fecaca; }
.risk-emergency { background: #dc2626; color: #fff; border: 1px solid #dc2626; }

/* Risk meter */
```

```
.risk-meter-wrap { padding: 16px 18px; border-bottom: 1px solid var(--border); }

.risk-meter-label {

  display: flex;

  justify-content: space-between;

  font-size: 12px;

  color: var(--muted);

  margin-bottom: 6px;

}

.risk-meter-track {

  height: 8px;

  background: #e5e7eb;

  border-radius: 4px;

  overflow: hidden;

}

.risk-meter-fill {

  height: 100%;

  border-radius: 4px;

  transition: width 0.8s ease;

}

.fill-low    { background: var(--success); }

.fill-moderate { background: var(--warning); }

.fill-high   { background: var(--danger); }

.fill-emergency { background: #7f1d1d; }

/* Sections */
```

```
.result-section { padding: 14px 18px; border-bottom: 1px solid var(--border); }
```

```
.result-section:last-child { border-bottom: none; }
```

```
.result-section-title {
```

```
  font-size: 11px;
```

```
  font-weight: 700;
```

```
  text-transform: uppercase;
```

```
  letter-spacing: 0.06em;
```

```
  color: var(--muted);
```

```
  margin-bottom: 10px;
```

```
  display: flex;
```

```
  align-items: center;
```

```
  gap: 6px;
```

```
}
```

```
.result-section-title::before {
```

```
  content: ";
```

```
  display: inline-block;
```

```
  width: 3px;
```

```
  height: 12px;
```

```
  border-radius: 2px;
```

```
  background: var(--accent);
```

```
}
```

```
/* Condition pills */
```

```
.conditions-list { display: flex; flex-direction: column; gap: 8px; }
```

```
.condition-item {
```

```
display: flex;

align-items: center;

justify-content: space-between;

padding: 9px 12px;

background: var(--surface);

border: 1px solid var(--border);

border-radius: 6px;

font-size: 13px;
}

.condition-name { font-weight: 600; color: var(--text); }

.condition-prob {

font-size: 11px;

font-weight: 700;

padding: 2px 8px;

border-radius: 12px;
}

.prob-high { background: #fee2e2; color: #b91c1c; }

.prob-mod { background: #fef9c3; color: #92400e; }

.prob-low { background: #dcfce7; color: #15803d; }


/* Recommendations */

.rec-list { display: flex; flex-direction: column; gap: 7px; }

.rec-item {

display: flex;

gap: 10px;
```

```
align-items: flex-start;

padding: 9px 12px;

background: var(--surface);

border-radius: 6px;

border-left: 3px solid var(--accent);

font-size: 13px;

color: var(--text);
}

.rec-item.urgent { border-left-color: var(--danger); background: #fef2f2; }
.rec-item.caution { border-left-color: var(--warning); background: #fffbeb; }
.rec-item.safe { border-left-color: var(--success); background: #f0fdf4; }
.rec-icon { font-size: 14px; flex-shrink: 0; margin-top: 1px; }
```

```
/* AI narrative */
```

```
.ai-narrative {

font-size: 13.5px;

line-height: 1.7;

color: var(--text);

white-space: pre-wrap;
}
```

```
/* Disclaimer */
```

```
.disclaimer {

background: #fffbeb;

border: 1px solid #fde68a;
```

```
border-radius: 6px;

padding: 10px 14px;

font-size: 12px;

color: #92400e;

display: flex;

gap: 8px;

align-items: flex-start;
}
```

```
/* — HISTORY — */
```

```
.history-list { display: flex; flex-direction: column; gap: 6px; }

.history-item {

padding: 10px 12px;

border: 1px solid var(--border);

border-radius: 6px;

cursor: pointer;

transition: all 0.13s;

background: var(--bg);

display: flex;

justify-content: space-between;

align-items: center;

}

.history-item:hover { border-color: var(--accent); background: #f0f7ff; }

.history-item-left { font-size: 12px; }

.history-item-symptoms { font-weight: 600; color: var(--text); margin-bottom: 2px; }
```

```
.history-item-time { color: var(--muted); font-size: 11px; }
```

```
.status-dot {  
  
  width: 8px; height: 8px;  
  
  border-radius: 50%;  
  
  flex-shrink: 0;  
}
```

```
.dot-low    { background: var(--success); }
```

```
.dot-moderate { background: var(--warning); }
```

```
.dot-high   { background: var(--danger); }
```

```
.dot-emergency { background: #7f1d1d; }
```

```
/* — SKELETON — */
```

```
.skeleton-wrap { padding: 18px; display: flex; flex-direction: column; gap: 12px; }
```

```
.skeleton {  
  
  background: linear-gradient(90deg, #e5e7eb 25%, #f3f4f6 50%, #e5e7eb 75%);  
  
  background-size: 200% 100%;  
  
  animation: shimmer 1.4s infinite;  
  
  border-radius: 6px;  
}
```

```
@keyframes shimmer { to { background-position: -200% 0; } }
```

```
.sk-line  { height: 14px; }
```

```
.sk-line-sm { height: 10px; width: 60%; }
```

```
.sk-block { height: 60px; }
```

```
/* — CONFIG — */
```

```
.config-toggle {  
  display: flex;  
  align-items: center;  
  gap: 8px;  
  font-size: 12px;  
  color: var(--muted);  
  cursor: pointer;  
  padding: 8px 18px;  
  border-top: 1px solid var(--border);  
  transition: color 0.13s;  
}
```

```
.config-toggle:hover { color: var(--text); }
```

```
.config-panel {  
  padding: 14px 18px;  
  border-top: 1px solid var(--border);  
  background: var(--surface);  
  display: none;  
  flex-direction: column;  
  gap: 10px;  
}
```

```
.config-panel.open { display: flex; }
```

```
/* — ERROR — */
```

```
.error-box {
```

```
background: #fef2f2;

border: 1px solid #fecaca;

border-radius: 6px;

padding: 14px 16px;

color: #b91c1c;

font-size: 13px;

display: flex;

gap: 10px;

align-items: flex-start;

}

.error-box-icon { font-size: 16px; flex-shrink: 0; }
```

```
/* — EMPTY — */
```

```
.empty-state {

text-align: center;

padding: 20px;

color: var(--muted);

font-size: 12.5px;

}
```

```
/* — FOOTER — */
```

```
footer {

text-align: center;

font-size: 12px;

color: #adb5bd;
```

```
border-top: 1px solid #e5e7eb;

padding: 16px;

margin-top: 24px;
}
</style>
</head>
<body>

<!-- HEADER -->
<header class="header">
  <div class="header-brand">

    <svg width="24" height="24" viewBox="0 0 24 24" fill="none">

      <circle cx="12" cy="12" r="11" stroke="#3b82d4" stroke-width="1.5"/>

      <path d="M12 7v5l3 3" stroke="#3b82d4" stroke-width="2" stroke-linecap="round"/>

      <path d="M8 12h1.5M14.5 12H16" stroke="#7c5cd8" stroke-width="1.5" stroke-linecap="round"/>

    </svg>

    AI Health Symptom Assessment Agent

    <span class="header-badge">IBM WatsonX</span>

  </div>

  <div class="header-meta">

    Powered by IBM Granite 3 &middot; Project c7ab2667

  </div>
</header>

<!-- LAYOUT -->
```

```
<div class="layout">
```

```
<!-- LEFT: INPUT -->
```

```
<aside>
```

```
<div class="card">
```

```
<div class="card-header">
```

```
<span class="card-header-dot" style="background:#3b82d4"></span>
```

```
Patient Information
```

```
</div>
```

```
<div class="card-body">
```

```
<div class="form-grid-2">
```

```
<div class="form-row">
```

```
<label for="age">Age</label>
```

```
<input type="number" id="age" min="1" max="120" placeholder="e.g. 35" />
```

```
</div>
```

```
<div class="form-row">
```

```
<label for="gender">Sex</label>
```

```
<select id="gender">
```

```
<option value="">Select</option>
```

```
<option>Male</option>
```

```
<option>Female</option>
```

```
<option>Other / Prefer not to say</option>
```

```
</select>
```

```
</div>
```

</div>

<div class="form-row">

<label for="duration">Symptom Duration</label>

<select id="duration">

<option value="">Select duration</option>

<option>Less than 24 hours</option>

<option>1–3 days</option>

<option>4–7 days</option>

<option>1–2 weeks</option>

<option>2–4 weeks</option>

<option>More than 1 month</option>

<option>Chronic / ongoing</option>

</select>

</div>

<div class="form-row">

<label for="severity">Overall Severity</label>

<select id="severity">

<option value="">Select severity</option>

<option>Mild – manageable, not interfering with daily activities</option>

<option>Moderate – somewhat limiting daily activities</option>

<option>Severe – significantly limiting daily activities</option>

<option>Very Severe – unable to perform daily activities</option>

</select>

</div>

<div class="form-row">

<label>Symptoms (add each symptom)</label>

<div class="tag-input-wrap">

<input type="text" id="symptomInput" placeholder="Type a symptom..." />

<button class="btn-tag-add" id="addSymptomBtn" title="Add symptom">+</button>

</div>

<div class="tags-container" id="tagsContainer"></div>

<div class="quick-symptoms" id="quickSymptoms"></div>

</div>

<div class="form-row">

<label for="medHistory">Relevant Medical History</label>

<textarea id="medHistory" placeholder="e.g. diabetes, hypertension, recent surgery, allergies..."></textarea>

</div>

<div class="form-row">

<label for="medications">Current Medications</label>

<input type="text" id="medications" placeholder="e.g. metformin, lisinopril, aspirin..." />

</div>

<button class="btn btn-primary" id="assessBtn">


```
<svg width="14" height="14" viewBox="0 0 24 24" fill="none" style="vertical-align:middle">
  <path d="M9 12l2 2 4-4" stroke="#fff" stroke-width="2" stroke-linecap="round" stroke-
linejoin="round"/>
```

```
<circle cx="12" cy="12" r="10" stroke="#fff" stroke-width="2"/>
```

```
</svg>
```

Assess Symptoms

```
</span>
```

```
<span class="spinner" id="spinner"></span>
```

```
</button>
```

```
<div class="btn-row">
```

```
<button class="btn btn-ghost" id="clearBtn" style="font-size:12px">Clear Form</button>
```

```
<button class="btn btn-ghost" id="exampleBtn" style="font-size:12px">Load Example</button>
```

```
</div>
```

```
</div>
```

```
<!-- CONFIG TOGGLE -->
```

```
<div class="config-toggle" id="configToggle">
```

```
<svg width="12" height="12" viewBox="0 0 24 24" fill="none">
```

```
<path d="M12 2a10 10 0 100 20A10 10 0 0012 2zm0 14v-4m0-4h.01" stroke="currentColor"
stroke-width="2" stroke-linecap="round"/>
```

```
</svg>
```

API Configuration

```
<svg id="configArrow" width="10" height="10" viewBox="0 0 24 24" fill="none" style="margin-
left:auto">
```

```
<path d="M6 9l6 6 6-6" stroke="currentColor" stroke-width="2" stroke-linecap="round"/>
```

```
</svg>
```

```

</div>

<div class="config-panel" id="configPanel">

  <div class="form-row" style="margin:0">

    <label>WatsonX API Key</label>

    <input type="text" id="apiKeyInput" value="0uyScL4GoQJAJMoWtG51xz3IsHtLZz_tvW4dD86s4Ab-"
  />

  </div>

  <!-- ID mode toggle -->

  <div class="form-row" style="margin:0">

    <label>ID Type</label>

    <div style="display:flex;gap:0;border:1px solid var(--border);border-radius:6px;overflow:hidden">

      <button id="btnUseProject" onclick="switchIdMode('project')"

        style="flex:1;padding:8px;border:none;cursor:pointer;font-size:12px;font-
weight:600;background:var(--accent);color:#fff;font-family:inherit">

        Project ID

      </button>

      <button id="btnUseSpace" onclick="switchIdMode('space')"

        style="flex:1;padding:8px;border:none;cursor:pointer;font-size:12px;font-
weight:600;background:var(--surface);color:var(--muted);font-family:inherit">

        Space ID

      </button>

    </div>

  </div>

  </div>

  <div class="form-row" style="margin:0" id="projectIdRow">

```

```

    <label>Project ID <span style="font-weight:400;text-transform:none;font-size:10px;color:#e67e00">(needs WML associated)</span></label>

    <input type="text" id="projectIdInput" value="c7ab2667-ee82-43eb-8bd9-b11a47a6ad8e" />

</div>

<div class="form-row" style="margin:0;display:none" id="spaceIdRow">

    <label>Space ID <span style="font-weight:400;text-transform:none;font-size:10px;color:var(--success)">✓ recommended</span></label>

    <input type="text" id="spaceIdInput" placeholder="Paste your Deployment Space ID..." />

</div>

<div class="form-row" style="margin:0">

    <label>Endpoint</label>

    <input type="text" id="endpointInput" value="https://us-south.ml.cloud.ibm.com" />

</div>

<div class="form-row" style="margin:0">

    <label>Model ID</label>

    <input type="text" id="modelIdInput" value="ibm/granite-3-8b-instruct" />

</div>

<!-- Help note -->

<div style="background:#ffffbeb;border:1px solid #fde68a;border-radius:6px;padding:10px 12px;font-size:11.5px;color:#92400e;line-height:1.5">

    <strong>403 no_associated_service_instance?</strong><br/>

    Your Project must have a <strong>Watson Machine Learning</strong> service associated.<br/>

    Go to <strong>IBM Cloud → WatsonX → your project → Manage → Services & integrations</strong> and add WML.<br/>

```

Or create a **Deployment Space** with WML and use **Space ID** above.

</div>

</div>

</div>

<!-- HISTORY -->

<div class="card" style="margin-top:16px">

<div class="card-header">

Assessment History

</div>

<div class="card-body" style="padding:12px">

<div id="historyList">

<div class="empty-state">No assessments yet.
Run your first assessment to see history here.</div>

</div>

</div>

</div>

</aside>

<!-- RIGHT: RESULTS -->

<main>

<div class="chat-panel" id="chatPanel">

<div class="welcome-box" id="welcomeBox">

```
<div class="welcome-icon">

  <svg width="28" height="28" viewBox="0 0 24 24" fill="none">

    <path d="M12 2C6.48 2 2 6.48 2 12s4.48 10 10 10 10-4.48 10-10S17.52 2 12 2z"
fill="rgba(255,255,255,0.2)"/>

    <path d="M12 7v5l3 3" stroke="#fff" stroke-width="2" stroke-linecap="round"/>

    <path d="M8.5 11h1M14.5 11h1" stroke="#fff" stroke-width="1.5" stroke-linecap="round"/>

  </svg>

</div>

<h2>Health Symptom Assessment Agent</h2>

<p>Powered by IBM Granite 3 on WatsonX. Enter your symptoms and patient information on the
left to receive an AI-powered health assessment with risk analysis and care recommendations.</p>

<div class="feature-grid">

  <div class="feature-item">

    <span class="feature-item-icon" style="background:#3b82d4">AI</span>

    <span>AI-driven symptom analysis using IBM Granite 3 LLM</span>

  </div>

  <div class="feature-item">

    <span class="feature-item-icon" style="background:#16a34a">Rx</span>

    <span>Personalised care recommendations and next steps</span>

  </div>

  <div class="feature-item">

    <span class="feature-item-icon" style="background:#d97706">R</span>

    <span>Risk stratification: Low, Moderate, High, Emergency</span>

  </div>

  <div class="feature-item">

    <span class="feature-item-icon" style="background:#7c5cd8">Dx</span>
```

```
        <span>Possible condition identification with probability scoring</span>
    </div>
</div>
</div>

<div class="error-box" id="errorBox" style="display:none">
    <span class="error-box-icon">&#9888;</span>
    <div id="errorMsg">An error occurred. Please check your API configuration and try again.</div>
</div>

<div id="resultArea"></div>
</div>
</main>
</div>

<footer>Made with IBM Bob</footer>

<script>
/* — STATE — */
let symptoms = [];
let assessmentHistory = [];

const QUICK = [
    'Headache','Fever','Cough','Fatigue','Nausea',
    'Chest pain','Shortness of breath','Dizziness','Back pain',
```

```
'Sore throat','Abdominal pain','Joint pain','Rash','Vomiting'  
];
```

```
/* — INIT — */
```

```
(function init() {  
  const qc = document.getElementById('quickSymptoms');  
  QUICK.forEach(s => {  
    const b = document.createElement('button');  
    b.className = 'quick-tag';  
    b.textContent = s;  
    b.addEventListener('click', () => addSymptom(s));  
    qc.appendChild(b);  
  });  
})();
```

```
/* — TAGS — */
```

```
function addSymptom(val) {  
  val = (val || document.getElementById('symptomInput').value).trim();  
  if (!val || symptoms.includes(val.toLowerCase())) return;  
  symptoms.push(val.toLowerCase());  
  renderTags();  
  document.getElementById('symptomInput').value = '';  
}  
  
function removeSymptom(idx) { symptoms.splice(idx, 1); renderTags(); }  
  
function renderTags() {
```

```

const c = document.getElementById('tagsContainer');

c.innerHTML = '';

symptoms.forEach((s, i) => {

  const span = document.createElement('span');

  span.className = 'tag';

  span.innerHTML = s + '<button class="tag-remove" title="Remove">&times;</button>';

  span.querySelector('.tag-remove').addEventListener('click', () => removeSymptom(i));

  c.appendChild(span);

});

}

document.getElementById('addSymptomBtn').addEventListener('click', () => addSymptom());

document.getElementById('symptomInput').addEventListener('keydown', e => {

  if (e.key === 'Enter') { e.preventDefault(); addSymptom(); }

});

/* — CONFIG TOGGLE — */

document.getElementById('configToggle').addEventListener('click', function() {

  const panel = document.getElementById('configPanel');

  panel.classList.toggle('open');

  document.getElementById('configArrow').style.transform =

    panel.classList.contains('open') ? 'rotate(180deg)' : '';

});

/* — CLEAR / EXAMPLE — */

document.getElementById('clearBtn').addEventListener('click', function() {

```

```

symptoms = []; renderTags();

['age','gender','duration','severity','medHistory','medications']
  .forEach(id => { document.getElementById(id).value = ''; });

hideError();

});

document.getElementById('exampleBtn').addEventListener('click', function() {

  document.getElementById('age').value = '45';

  document.getElementById('gender').value = 'Male';

  document.getElementById('duration').value = '4–7 days';

  document.getElementById('severity').value = 'Moderate – somewhat limiting daily activities';

  document.getElementById('medHistory').value = 'Hypertension, Type 2 Diabetes';

  document.getElementById('medications').value = 'Metformin 500mg, Lisinopril 10mg';

  symptoms = [];

  ['chest pain','shortness of breath','fatigue','sweating','mild nausea'].forEach(addSymptom);

});

/* — IBM IAM TOKEN — */

async function getIBMTOKEN(apiKey) {

  // Route through local proxy to avoid CORS when served via http://localhost

  const resp = await fetch('/proxy/iam', {

    method: 'POST',

    headers: { 'Content-Type': 'application/x-www-form-urlencoded' },

    body: 'grant_type=urn%3Aibm%3Aparams%3Aoauth%3Agrant-type%3Aapikey&apikey=' +
    encodeURIComponent(apiKey)

  });

  if (!resp.ok) throw new Error('Failed to obtain IBM IAM token. Please verify your API key.');
```

```
const data = await resp.json();

return data.access_token;

}
```

```
/* — PROMPT — */
```

```
function buildPrompt(age, gender, duration, severity, symptomList, medHistory, medications) {

  const sevLabel = severity.split(' - ')[0] || severity;

  return `You are a knowledgeable AI health assessment assistant. Analyze the following patient
information and reported symptoms. Provide a structured, thorough health assessment.
```

PATIENT PROFILE:

- Age: \${age || 'Not specified'}
- Sex: \${gender || 'Not specified'}
- Symptom Duration: \${duration || 'Not specified'}
- Severity: \${sevLabel || 'Not specified'}
- Reported Symptoms: \${symptomList.length ? symptomList.join(', ') : 'None reported'}
- Medical History: \${medHistory || 'None reported'}
- Current Medications: \${medications || 'None reported'}

INSTRUCTIONS:

Provide a structured health assessment in the following exact JSON format (respond ONLY with the JSON, no extra text):

```
{

  "risk_level": "Low|Moderate|High|Emergency",

  "risk_score": <integer 0-100>,


```

```

    "risk_explanation": "<1-2 sentence explanation of the risk level>",

    "possible_conditions": [

        {"name": "<condition name>", "probability": "High|Moderate|Low", "description": "<brie
description>"},

        {"name": "<condition name>", "probability": "High|Moderate|Low", "description": "<brie
description>"},

        {"name": "<condition name>", "probability": "Moderate|Low", "description": "<brie
description>"}

    ],

    "recommendations": [

        {"type": "urgent|caution|safe", "text": "<recommendation>"},

        {"type": "urgent|caution|safe", "text": "<recommendation>"},

        {"type": "urgent|caution|safe", "text": "<recommendation>"},

        {"type": "urgent|caution|safe", "text": "<recommendation>"}

    ],

    "assessment_summary": "<A clear, empathetic 3-4 sentence clinical summary of the assessment,
findings, and advice>",

    "red_flags": ["<red flag symptom or sign if any>"],

    "follow_up": "<specific follow-up recommendation with timeframe>"

}

```

Be thorough, clinically accurate, and empathetic. Always recommend professional medical consultation.`;

```

}

```

```

/* — ID MODE TOGGLE — */

```

```

let _idMode = 'project';

```

```

function switchIdMode(mode) {

```

```

_idMode = mode;

document.getElementById('projectIdRow').style.display = mode === 'project' ? '' : 'none';

document.getElementById('spaceIdRow').style.display = mode === 'space' ? '' : 'none';

document.getElementById('btnUseProject').style.background = mode === 'project' ? 'var(--accent)' :
'var(--surface)';

document.getElementById('btnUseProject').style.color = mode === 'project' ? '#fff' : 'var(--muted)';

document.getElementById('btnUseSpace').style.background = mode === 'space' ? 'var(--accent)' :
'var(--surface)';

document.getElementById('btnUseSpace').style.color = mode === 'space' ? '#fff' : 'var(--muted)';
}

```

```

/* — CALL WATSONX — */

```

```

async function callWatsonX(prompt) {

  const apiKey = document.getElementById('apiKeyInput').value.trim();

  const modelId = document.getElementById('modelIdInput').value.trim();

  const token = await getIBMTOKEN(apiKey);

  // Use space_id or project_id depending on toggle

  const idKey = _idMode === 'space' ? 'space_id' : 'project_id';

  const idVal = _idMode === 'space'

    ? document.getElementById('spaceIdInput').value.trim()

    : document.getElementById('projectIdInput').value.trim();

  if (!idVal) throw new Error('Please enter a ' + (idVal === 'space_id' ? 'Space ID' : 'Project ID') + ' in API
  Configuration.');
```

```
const body = {  
  model_id: modelId,  
  [idKey]: idVal,  
  messages: [{ role: 'user', content: prompt }],  
  parameters: {  
    max_new_tokens: 1200,  
    temperature: 0.3,  
    top_p: 0.9,  
    repetition_penalty: 1.05  
  }  
};
```

```
// Route through local proxy to avoid CORS when served via http://localhost
```

```
const resp = await fetch('/proxy/watsonx', {  
  method: 'POST',  
  headers: {  
    'Authorization': 'Bearer ' + token,  
    'Content-Type': 'application/json',  
    'Accept': 'application/json'  
  },  
  body: JSON.stringify(body)  
});
```

```
if (!resp.ok) {  
  const errText = await resp.text();
```

```

    throw new Error('WatsonX API error ' + resp.status + ': ' + errText.slice(0, 300));
  }

  const data = await resp.json();

  return (data?.choices?.[0]?.message?.content || data?.results?.[0]?.generated_text || '').trim();
}

/* — PARSE — */

function parseAssessment(raw) {
  const match = raw.match(/\{[\s\S]*\}/);

  if (!match) throw new Error('Could not parse assessment response. The model returned an unexpected format.');

  return JSON.parse(match[0]);
}

/* — HELPERS — */

function riskClass(l) { l=(l||'').toLowerCase(); return l==='emergency'?'risk-emergency':l==='high'?'risk-high':l==='moderate'?'risk-moderate':'risk-low'; }

function fillClass(l) { l=(l||'').toLowerCase(); return l==='emergency'?'fill-emergency':l==='high'?'fill-high':l==='moderate'?'fill-moderate':'fill-low'; }

function dotClass(l) { l=(l||'').toLowerCase(); return l==='emergency'?'dot-emergency':l==='high'?'dot-high':l==='moderate'?'dot-moderate':'dot-low'; }

function probClass(p) { p=(p||'').toLowerCase(); return p==='high'?'prob-high':p==='moderate'?'prob-mod':'prob-low'; }

function recClass(t) { t=(t||'').toLowerCase(); return t==='urgent'?'urgent':t==='caution'?'caution':'safe'; }

function recIcon(t) { t=(t||'').toLowerCase(); return t==='urgent'?'⚠️':t==='caution'?'⚠️':t==='safe'?'✅'; }

```

```
function escHtml(str) { return
String(str).replace(/&/g,'&amp;').replace(/</g,'&lt;').replace(/>/g,'&gt;').replace(/"/g,'&quot;'); }
```

```
/* — RENDER RESULT — */
```

```
function renderResult(assessment, ts) {

  const risk = assessment.risk_level || 'Unknown';

  const score = Math.min(100, Math.max(0, assessment.risk_score || 0));

  const uid = 'fill_' + Date.now();

  const conds = assessment.possible_conditions || [];

  const recs = assessment.recommendations || [];

  const flags = assessment.red_flags || [];
```

```
let html = `

<div class="assessment-card">

  <div class="assessment-header">

    <div class="assessment-title">

      <svg width="16" height="16" viewBox="0 0 24 24" fill="none">

        <path d="M9 12l2 2 4-4M12 2a10 10 0 100 20A10 10 0 0012 2z" stroke="var(--accent)" stroke-
width="2" stroke-linecap="round"/>

      </svg>

      Assessment Report

    </div>

    <div style="display:flex;align-items:center;gap:10px">

      <span class="risk-badge ${riskClass(risk)}">${escHtml(risk)} Risk</span>

      <span class="assessment-timestamp">${ts}</span>

    </div>

  </div>

</div>
```

</div>

<div class="risk-meter-wrap">

<div class="risk-meter-label">

Risk Score

\${score}/100

</div>

<div class="risk-meter-track">

<div class="risk-meter-fill \${fillClass(risk)}" id="\${uid}" style="width:0%"></div>

</div>

<div style="font-size:11.5px;color:var(--muted);margin-top:6px">\${escHtml(assessment.risk_explanation | | "")}</div>

</div>`;

if (assessment.assessment_summary) {

html += `<div class="result-section"><div class="result-section-title">Clinical Summary</div>

<div class="ai-narrative">\${escHtml(assessment.assessment_summary)}</div></div>`;

}

if (conds.length) {

html += `<div class="result-section"><div class="result-section-title">Possible Conditions</div>

<div class="conditions-list">`;

conds.forEach(c => {

html += `<div class="condition-item">

<div>

<div class="condition-name">\${escHtml(c.name | | "")}</div>

```
    <div style="font-size:11.5px;color:var(--muted);margin-top:2px">${escHtml(c.description | | "")}</div>
```

```
  </div>
```

```
    <span class="condition-prob ${probClass(c.probability)}">${escHtml(c.probability | | "")}</span>
```

```
  </div>`;
```

```
});
```

```
html += `</div></div>`;
```

```
}
```

```
if (flags.length) {
```

```
  html += `<div class="result-section">
```

```
    <div class="result-section-title" style="color:#b91c1c">&#9888; Red Flag Symptoms Noted</div>
```

```
    <div style="display:flex;flex-wrap:wrap;gap:6px">
```

```
      ${flags.map(f => `<span class="risk-badgerisk-high">${escHtml(f)}</span>`).join("")}
```

```
    </div></div>`;
```

```
}
```

```
if (recs.length) {
```

```
  html += `<div class="result-section"><div class="result-section-title">Recommendations</div>
```

```
    <div class="rec-list">`;
```

```
  recs.forEach(r => {
```

```
    html += `<div class="rec-item ${recClass(r.type)}">
```

```
      <span class="rec-icon">${recIcon(r.type)}</span>
```

```
      <span>${escHtml(r.text | | "")}</span>
```

```
    </div>`;
```

```
  });
```

```

    html += `</div></div>`;

}

if (assessment.follow_up) {

    html += `<div class="result-section"><div class="result-section-title">Follow-up Plan</div>

    <div style="font-size:13px;color:var(--text)">${escHtml(assessment.follow_up)}</div></div>`;

}

html += `<div class="result-section">

    <div class="disclaimer">

        <span style="font-size:16px;flex-shrink:0">&#9432;</span>

        <span><strong>Medical Disclaimer:</strong> This AI-generated assessment is for informational
        purposes only and does not constitute professional medical advice, diagnosis, or treatment. Always
        consult a qualified healthcare professional for medical concerns.</span>

    </div></div>

</div>`;

return { html, uid, score, risk };

}

/* — ASSESS — */

document.getElementById('assessBtn').addEventListener('click', async function() {

    if (!symptoms.length) { showError('Please add at least one symptom before running the assessment.');
```

return; }

```

    hideError();

    setLoading(true);

    document.getElementById('welcomeBox').style.display = 'none';

```

```
document.getElementById('resultArea').innerHTML = loadingSkeleton();
```

```
try {
```

```
    const age    = document.getElementById('age').value.trim();
```

```
    const gender  = document.getElementById('gender').value;
```

```
    const duration = document.getElementById('duration').value;
```

```
    const severity = document.getElementById('severity').value;
```

```
    const medHistory = document.getElementById('medHistory').value.trim();
```

```
    const medications = document.getElementById('medications').value.trim();
```

```
    const prompt = buildPrompt(age, gender, duration, severity, symptoms, medHistory, medications);
```

```
    const raw = await callWatsonX(prompt);
```

```
    const assessment = parseAssessment(raw);
```

```
    const ts = new Date().toLocaleString('en-US', { month:'short', day:'numeric', hour:'2-digit', minute:'2-digit' });
```

```
    const { html, uid, score, risk } = renderResult(assessment, ts);
```

```
    document.getElementById('resultArea').innerHTML = html;
```

```
    setTimeout(() => { const f = document.getElementById(uid); if (f) f.style.width = score + '%'; }, 100);
```

```
    assessmentHistory.unshift({
```

```
        ts,
```

```
        symptomsLabel: symptoms.slice(0,3).join(', ') + (symptoms.length > 3 ? '...' : ''),
```

```
        risk, score, html, uid
```

```
    });
```

```

renderHistory();

} catch (err) {

  document.getElementById('resultArea').innerHTML = "";

  showError(err.message || 'An unexpected error occurred.');
```

}

```

  finally {

    setLoading(false);

  }

});

/* — HISTORY — */

function renderHistory() {

  const list = document.getElementById('historyList');

  if (!assessmentHistory.length) { list.innerHTML = '<div class="empty-state">No assessments yet.</div>';
  return; }

  list.innerHTML = assessmentHistory.map((e, i) => `

    <div class="history-item" data-idx="${i}">

      <div class="history-item-left">

        <div class="history-item-symptoms">${escHtml(e.symptomsLabel)}</div>

        <div class="history-item-time">${e.ts} &middot; Score: ${e.score}/100</div>

      </div>

      <span class="status-dot ${dotClass(e.risk)}" title="${escHtml(e.risk)} Risk"></span>

    </div>` ).join("");

  list.querySelectorAll('.history-item').forEach(el => {

    el.addEventListener('click', function() {

      const entry = assessmentHistory[parseInt(this.dataset.idx)];
```

```

    document.getElementById('welcomeBox').style.display = 'none';

    document.getElementById('resultArea').innerHTML = entry.html;

    setTimeout(() => { const f = document.getElementById(entry.uid); if (f) f.style.width = entry.score +
    '%'; }, 100);

    });

    });

}

```

```

/* — UI HELPERS — */

```

```

function setLoading(on) {

    const btn = document.getElementById('assessBtn');

    btn.disabled = on;

    btn.classList.toggle('loading', on);

    document.getElementById('spinner').style.display = on ? 'block' : 'none';

}

function showError(msg) {

    document.getElementById('errorMsg').textContent = msg;

    document.getElementById('errorBox').style.display = 'flex';

}

function hideError() { document.getElementById('errorBox').style.display = 'none'; }

function loadingSkeleton() {

    return `<div class="card"><div class="skeleton-wrap">

        <div class="skeleton sk-line" style="width:50%"></div>

        <div class="skeleton sk-block"></div>

        <div class="skeleton sk-line"></div>

        <div class="skeleton sk-line-sm"></div>
    `;
}

```

```
<div class="skeleton sk-block"></div>

<div class="skeleton sk-line"></div>

<div class="skeleton sk-line-sm"></div>

</div></div>`;

}

</script>

</body>

</html>
```